

Defect Type Catalog — Quality Reference

E-Bike Manufacturing — Enhanced MES Reference

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This catalog enumerates every defect type recorded in the MES (**1,549 findings** across **38 types**), grouped by the production area where it is most commonly raised. Each defect carries a severity of 1 (minor) to 5 (critical), a positional location, a root-cause category and a disposition. Counts and severity ranges below are computed directly from the Defects table.

1. Severity Scale

Severity	Label	Handling
1	Minor (327)	Cosmetic / accept or touch-up
2	Low (304)	Minor rework at station
3	Moderate (299)	Rework, re-inspect before release
4	Major (325)	Major — quarantine, supervisor review
5	Critical (294)	Critical — stop, RCA, likely scrap

2. Defect Types by Area

Frame Welding

Defect Type	Count	Severity range	Typical
Misalignment	76	1–5	Moderate
Dimension Error	74	1–5	Moderate
Weld Failure	72	1–5	Moderate
Structural Weakness	65	1–5	Moderate

Battery Assembly

Defect Type	Count	Severity range	Typical
Battery Capacity	68	1–5	Moderate
Connection Issue	68	1–5	Moderate

Motor Assembly

Defect Type	Count	Severity range	Typical
Sensor Malfunction	65	1–5	Moderate
Motor Output	63	1–5	Moderate

Wheel Building

Defect Type	Count	Severity range	Typical
Bead Seating Issue	42	1–5	Low
Hub Play	35	1–5	Moderate
Spoke Tension	29	1–5	Moderate
Valve Stem Defect	25	1–5	Moderate
Out of True	24	1–5	Moderate

Paint Booth

Defect Type	Count	Severity range	Typical
Surface Defect	76	1–5	Moderate
Drips	4	2–4	Moderate
Color Mismatch	4	2–4	Moderate
Orange Peel	3	2–3	Moderate
Poor Adhesion	2	2	Low
Uneven Coating	1	2	Low

Final Assembly / Drivetrain / Brakes

Defect Type	Count	Severity range	Typical
Cable Routing	141	1–5	Moderate
Interface Fit	139	1–5	Moderate
Adjustment Error	135	1–5	Moderate
Loose Fastener	122	1–5	Moderate
Missing Component	115	1–5	Moderate
Display Fault	75	1–5	Moderate
Bottom Bracket Noise	3	4	Major
Vibration	3	2–4	Moderate
Derailleur Adjustment	2	3	Moderate
Lever Play	2	2–4	Moderate
Pad Wear	2	4	Major
Rotor Warp	2	3	Moderate
Chain Misalignment	1	4	Major
Crank Play	1	3	Moderate
Hydraulic Leak	1	5	Critical
Noise	1	4	Major

Packaging / General

Defect Type	Count	Severity range	Typical
Packaging Damage	4	3–5	Major
Documentation Error	3	2–5	Major
Cosmetic Damage	1	1	Minor

3. Root-Cause Categories

Every defect is assigned one of six root-cause categories. Distribution across all findings:

Root Cause	Findings	Primary countermeasure
Process Variation	276	Drift within nominal process limits — addressed with SPC and parameter control.
Machine Calibration	264	Equipment out of calibration — addressed via the calibration schedule.
Material Defect	262	Incoming component non-conformance — addressed with supplier quality and incoming inspection.
Design Issue	260	Root cause in product or fixture design — routed to engineering change.
Operator Error	245	Handling, assembly or setup mistakes — addressed through training and poka-yoke.
Tooling Wear	242	Worn tips, electrodes, fixtures — addressed by tool-life tracking and replacement.

4. Dispositions

The action taken on each defect is recorded in Defects.ActionTaken:

Action	Count	Meaning
Returned to Supplier	284	Non-conforming material sent back to vendor.
Scrapped	282	Unit or component discarded.
Reworked	249	Corrected in-line and re-inspected.
Repaired	248	Fixed off-line, returned to flow.
Process Adjusted	248	Process parameters corrected to remove the cause.
Accepted with Deviation	238	Accepted under documented concession.

5. Location Coding

Defect location is captured as a positional zone on the unit: Bottom (206), Center (223), Front (198), Left (207), Rear (242), Right (233), Top (240).

Root-cause references in this catalog map directly to the Root Cause Analysis Guide, which gives investigation and corrective steps per failure mode.